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Kinetics and Reactor Design (CHE F311)

Objectives

- Examine the effects of heat in chemical reactors
 - The basic design equations, rate laws and stoichiometric relationships derived and used for isothermal reactors are still valid for non-isothermal reactors
 - The major difference in non-isothermal reactors is that the temperature varies along the length of the PFR or heat is transferred to or from the CSTR

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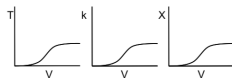
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Kinetics and Reactor Design (CHE F311)

Introduction

Let's calculate the volume necessary to achieve a conversion, X , in a PFR for a $A \rightarrow B$ first-order, exothermic and adiabatic reaction.

The temperature profile might look something like this:



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Performance Equations

Mole Balance: $\frac{dX}{dV} = \frac{-r_A}{F_{A0}}$

Rate Law: $r_A = -k_1 \exp\left[\frac{E}{R}\left(\frac{1}{T_1} - \frac{1}{T}\right)\right] C_A$

Stoichiometry: $C_A = C_{A0}(1-X)$

Combine: $\frac{dX}{dV} = \frac{k_1 \exp\left[\frac{E}{R}\left(\frac{1}{T_1} - \frac{1}{T}\right)\right] C_{A0}(1-X)}{F_{A0}}$

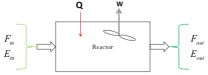
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Energy Balance



$$\left(\begin{array}{c} \text{Rate of accumulation of energy within the system} \\ \frac{d\hat{E}_{tot}}{dt} \end{array} \right) = \left(\begin{array}{c} \text{Rate of flow of heat to the system from the surroundings} \\ \dot{Q} \end{array} \right) - \left(\begin{array}{c} \text{Rate of work done by the system on the surroundings} \\ \dot{W} \end{array} \right) + \left(\begin{array}{c} \text{Rate of energy added to the system by mass flow into the system} \\ F_{in} E_{in} \end{array} \right) - \left(\begin{array}{c} \text{Rate of energy leaving system by mass flow out of the system} \\ F_{out} E_{out} \end{array} \right)$$

$$\frac{d\hat{E}_{tot}}{dt} \text{ (J/s)} = \dot{Q} \text{ (J/s)} - \dot{W} \text{ (J/s)} + F_{in} E_{in} \text{ (J/s)} - F_{out} E_{out} \text{ (J/s)}$$

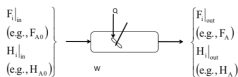
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Kinetics and Reactor Design (CHE 7302)

Energy Balance



Energy Balance on an open system: schematic.

$$\dot{Q} - \dot{W}_k + \sum F_{in} E_{in} - \sum F_{out} E_{out} = \frac{dE_{tot,sys}}{dt}$$

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Kinetics and Reactor Design (CHE 7302)

Assumptions

$E_i = U_i + P_i E_i^{s0} + K E_i^{s0}$ Other energies small compared to internal
 $\dot{W}$ = flow work + shaft work

$$\text{flow work} = - \sum F_{in} P_{in} \bar{V}_{in} + \sum F_{out} P_{out} \bar{V}_{out} \quad \left(\bar{V} = \frac{m^3}{\text{mol}} \right)$$

$$\sum F_{in} U_{in} - \sum F_{out} U_{out} + \dot{Q} - \left[- \sum F_{in} P_{in} \bar{V}_{in} + \sum F_{out} P_{out} \bar{V}_{out} \right] + \dot{W}_s = \frac{dE_{tot,sys}}{dt}$$

$$\sum F_{in} \left[U_{in} + P_{in} \bar{V}_{in} \right] - \sum F_{out} \left[U_{out} + P_{out} \bar{V}_{out} \right] + \dot{Q} - \dot{W}_s = \frac{dE_{tot,sys}}{dt}$$

$$\sum F_{in} H_{in} - \sum F_{out} H_{out} + \dot{Q} - \dot{W}_s = \frac{dE_{tot,sys}}{dt}$$

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Heat Effects

$$\sum F_{i0} H_{i0} - F_{i0} \sum \Theta_i H_{i0}$$

$$\sum F_i H_i = F_{i0} \sum (\Theta_i + v_i X) H_i = F_{i0} \sum \Theta_i H_i - F_{i0} X \sum v_i H_i$$

$$\dot{Q} - \dot{W}_s + F_{i0} (\sum \Theta_i (H_{i0} - H_i)) - F_{i0} X \Delta H_{R0} = 0$$

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For No Phase Changes

Neglect any enthalpy changes resulting from the mixing $\rightarrow$ partial molar enthalpies pure comp

$$H_i(T) = H_i^0(T_R) + \int_{T_R}^T C_{p,i} dT$$

$\hookrightarrow$ Enthalpy of formation at temperature T_R

Constant Heat Capacities

$$\rightarrow H_i(T) = H_i^0(T_R) + C_{p,i}(T - T_R)$$

$$\sum v_i H_i = \sum v_i H_i^0 + \sum v_i C_{p,i}(T - T_R)$$

 $\hookrightarrow$ Heat of reaction at temperature T

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$$\sum v_i H_i = \sum v_i H_i^0 + \sum v_i C_{p,i}(T - T_R)$$

$$\Delta H_R(T) = \Delta H_R^0(T_R) + \Delta \hat{C}_p(T - T_R)$$

$$\sum v_i \hat{C}_{p,i} = \Delta \hat{C}_p = \frac{d}{dT} \hat{C}_{p,D} + \frac{c}{dT} \hat{C}_{p,C} - \frac{b}{dT} \hat{C}_{p,A} - \hat{C}_{p,B}$$

Substituting back into the Energy Balance

$$\dot{Q} - \dot{W}_s - F_{i0} X [\Delta H_R^0(T_R) + \Delta \hat{C}_p(T - T_R)] - F_{i0} \sum \Theta_i \hat{C}_{p,i}(T - T_R) = 0$$

Adiabatic ($\dot{Q}=0$) and no work

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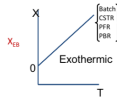
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Adiabatic Energy Balance

Adiabatic ($\dot{Q}=0$) and no work ($\dot{W}_s=0$)

$$X = - \frac{\sum \Theta_i \tilde{C}_{p,i}(T - T_R)}{[\Delta H_{R0}^0(T_R) + \Delta \hat{C}_p(T - T_R)]}$$



$$T = \frac{X[-\Delta H_{R0}^0(T_R)] + \sum \Theta_i C_{p,i} T_0 + X \Delta C_p T_R}{\sum \Theta_i C_{p,i} + X \Delta C_p}$$

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Kinetics and Reactor Design (CHE 7522)

Adiabatic PFR

- Normal butane C₄H₁₀, is to be isomerized to isobutene in a plug-flow reactor. Isobutene is a valuable product that is used in the manufacture of gasoline additives. The reaction is to be carried out adiabatically in the liquid phase under high pressure using essentially trace amounts of a liquid catalyst which gives a specific reaction rate of 31.1 h⁻¹ at 360 K. Calculate the PFR volume necessary to process 10⁵ gal/day (163 kmol/h) at 70% conversion of a mixture 90 mol % n-butane and 10 mol % i-pentane, which is considered an inert. The feed enters at 330 K.

Kinetics and Reactor Design (CHE 7522)

$$\Delta H_{Rx} = -6900 \text{ J/mol-butane, Activation energy} = 65.7 \text{ kJ/mol}$$

$$K_C = 3.03 \text{ at } 60^\circ\text{C, } C_{A0} = 9.3 \text{ kmol/dm}^3 = 9.3 \text{ kmol/m}^3$$

$$\text{Butane } C_{p,B} = 141 \text{ J/mol}\cdot\text{K} \quad \text{i-Pentane } C_{p,P} = 161 \text{ J/mol}\cdot\text{K}$$

$$C_{p,B} = 141 \text{ J/mol}\cdot\text{K} = 141 \text{ kJ/kmol}\cdot\text{K}$$



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Kinetics and Reactor Design (CHE 7522)

X	T (K)	k (h ⁻¹)	K _C	X _C	-r _A (kmol/m ³ ·h)	$\frac{F_{A0}}{-r_A}$ (m ³)
0	330	4.22	3.1	0.76	39.2	3.74
0.2	338.7	7.76	2.9	0.74	52.8	2.78
0.4	347.3	14.02	2.73	0.73	58.6	2.50
0.6	356.0	24.27	2.57	0.72	37.7	3.88
0.65	358.1	27.74	2.54	0.718	24.5	5.99
0.7	360.3	31.67	2.5	0.715	6.2	23.29

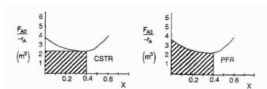
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Kinetics and Reactor Design (CHE 7522)

For 40% conversion



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Kinetics and Reactor Design (ENR 7522)

Adiabatic PFR (Eg. 8.3)

```
function dyfuncedV = ODEfun(V,Yfuncvec);
X = Yfuncvec(1);
% Explicit equations
Ca0 = 9.3;
Fa0 = 0.9 * 163;
T = 330 + 43.3 * X;
Kc = 3.03 * exp(-830.3 * (T - 333) / (T * 333));
k = 31.1 * exp(7906 * (T - 360) / (T * 360));
Xe = Kc / (1 + Kc);
ra = 0 - (k * Ca0 * (1 - ((1 + 1 / Kc) * X)));
rate = 0 - ra;
% Differential equation
dXdV = 0 - (ra / Fa0);
dyfuncedV = [dXdV];
```

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Kinetics and Reactor Design (ENR 7522)

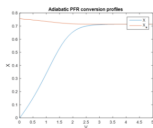
```
clc
Vspan = [0 5]; % Range for the independent variable i.e
volume of the reactor
y0 = [0]; % Initial values for the dependent variable i.e X
[V,y]=ode45(@ODEfun,Vspan,y0);
z=size(y);
for l=1:z(1,1)
T(l) = 330 + 43.3 * y(l,1);
Kc(l) = 3.03 * exp(-830.3 * (T(l) - 333) / (T(l) * 333));
Xe(l) = Kc(l) / (1 + Kc(l)); % Calculating the values of
equilibrium conversion from its expression
end
plot(V,y,V,Xe)
xlabel('V')
ylabel('X')
legend('X','X e')
title('Adiabatic PFR conversion profiles')
```

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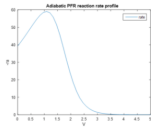
```
clc
Vspan = [0 5]; % Range for the independent variable i.e
volume of the reactor
y0 = [0]; % Initial values for the dependent variable i.e X
[V,y]=ode45(@ODEfun,Vspan,y0);
z=size(y);
Ca0 = 9.3;
for l=1:z(1,1)
T(l) = 330 + 43.3 * y(l,1);
Kc(l) = 3.03 * exp(-830.3 * (T(l) - 333) / (T(l) * 333));
k(l) = 31.1 * exp(7906 * (T(l) - 360) / (T(l) * 360));
ra(l) = - (k(l) * Ca0 * (1 - ((1 + 1 / Kc(l)) * y(l,1)))); %
Calculating the values of rate from its expression
end
plot(V,-ra)
xlabel('V')
ylabel('-ra')
legend('rate')
title('Adiabatic PFR reaction rate profile')
```

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Kinetics and Reactor Design (CHE 5302)



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Kinetics and Reactor Design (CHE 5302)



```

clear
Vspan = [0 5]; % Range for the independent
variable i.e volume of the reactor
y0 = [0]; % Initial values for the dependent
variable i.e x
[v,y]=ode45(@CODEfun,Vspan,y0);
z=size(y);
for i=1:z(1,1)
    T(i) = 330 + 43.3 * y(i,1); % Calculating the
    values of T from its expression
end
plot(v,T)
xlabel('V')
ylabel('T')
legend('T')
title('Adiabatic PFR temperature profile')

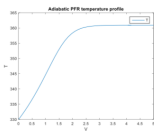
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Kinetics and Reactor Design (CHE 5302)



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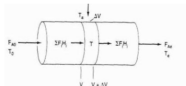
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Kinetics and Reactor Design (CHE 5302)



Steady-State Tubular Reactor with Heat Exchange

Assume that there are no radial gradients in the reactor



$$\frac{dT}{dV} = \frac{\text{Heat "Generation" Removed}}{F_{R0} C_{pR}} = \frac{-r_p \Delta H_{Rx} - U_a (T - T_c)}{F_{R0} C_{pR}}$$

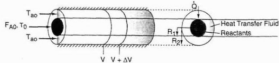
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Kinetics and Reactor Design (CHE 7302)

Balance on the Coolant Heat Transfer Fluid: Co-current



Heat added- Endo
Heat removed- Exo

If the flow rate of the heat transfer fluid is sufficiently high with respect to the heat released (or adsorbed) by the reacting mixture, then the heat transfer fluid temperature will be constant along the reactor

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Kinetics and Reactor Design (CHE 7302)

Production of Acetic Anhydride

In a treatment of the design of an acetic anhydride manufacturing facility, states that one of the key steps is the vapor-phase cracking of acetone to ketene and methane: $\text{CH}_3\text{COCH}_3 \rightarrow \text{CH}_2\text{CO} + \text{CH}_4$. The reaction is first order and is given by $\ln k = 34.34 - \frac{34222}{T}$ (T in K). In this design it is desired to feed 7850 kg of acetone per hour to a tubular reactor. The reactor consists of a bank of 1000 1-inch schedule 40 tubes. We will consider three cases:

CASE 1: The reactor is operated adiabatically:

CASE 2: The reactor is surrounded by a heat exchanger where the heat-transfer coefficient is $110 \text{ J/m}^2 \cdot \text{s} \cdot \text{K}$, and the temperature of the heating medium, T_a , is constant at 1150 K.

CASE 3: The heat exchanger in Case 2 now has a variable heating medium temperature

The inlet temperature and pressure are the same for both cases at 1035 K and 162 kPa (1.6 atm), respectively. Plot the conversion and temperature along the length of the reactor.

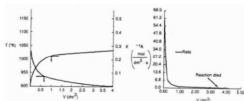
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Kinetics and Reactor Design (CHE 7302)

Case 1



As temperature drops, so does k and hence the rate, $-R_A$, drops to an insignificant value. Note that for this adiabatic endothermic reaction, the reaction virtually dies out after 3.5 m³, owing to the large drop in temperature, and very little conversion is achieved beyond this point.

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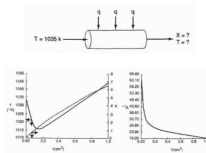
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Kinetics and Reactor Design (CHE 7302)

HEAT EXCHANGE WITH CONSTANT HEATING MEDIUM

TEMPERATURE



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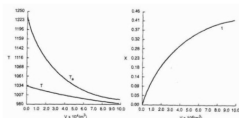
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Kinetics and Reactor Design (ENR F102)

HEAT EXCHANGE WITH VARIABLE HEATING MEDIUM

TEMPERATURE



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Kinetics and Reactor Design (ENR F102)

Equilibrium

Ethane cracking

- The thermal cracking of ethane is carried out in multitubular reactor. Typical production capacity of each tube is 10000 Tons per annum.
- Reactor specifications:
 - ↳ Feed to the reactor: ethane + steam (?)
 - ↳ Inlet pressure - 2.99 atm; temperature - 680°C
 - ↳ Tube length 95 m, ethylene conversion - 60%

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Kinetics and Reactor Design (ENR F102)

Thermodynamics

Considerations:

- Equilibrium conversion
- Conditions in the reactor
- Heat effects

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Kinetics and Reactor Design (ENR F102)

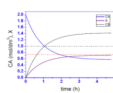
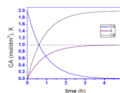
Thermodynamics?

A→B

$$r = 2.6 \times 10^6 e^{-\left(\frac{15000}{T}\right)} C_A$$

A↔B

$$r = 2.6 \times 10^6 e^{-\left(\frac{15000}{T}\right)} C_A - 3.9 \times 10^5 e^{-\left(\frac{12000}{T}\right)} C_B$$



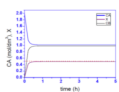
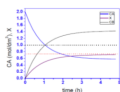
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Kinetics and Reactor Design (CHE F222)



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Kinetics and Reactor Design (CHE F222)



- Consider a reaction $\sum v_j A_j = 0$, at const T, P .
- The system will spontaneously change in the direction of increasing entropy, reaching equilibrium when entropy can not increase further.

- Free energy and Gibbs's equations

$$dG = Vdp - SdT + \sum_j \mu_j dn_j$$

$$\text{Chemical Potential } \mu_j = \left(\frac{\partial G}{\partial n_j} \right)_{T, P, n_i}$$

Amount by which the capacity of the phase to do the work by incremental addition of N_j

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Kinetics and Reactor Design (CHE F222)



Equilibrium

Gibb's Equation $dG = Vdp - SdT + \sum_j \mu_j dn_j$

Progress of reaction $n_j = n_{j0} + v_j \alpha$ or $dn_j = v_j d\alpha$

$$\Rightarrow dG = Vdp - SdT + \sum_j v_j \mu_j d\alpha$$

➤ Equilibrium condition

$$\left(\frac{\partial G}{\partial \alpha} \right)_{T, P} = \sum_j v_j \mu_j = 0$$

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Kinetics and Reactor Design (CHE F222)



Equilibrium

Gibb's Equation $dG = Vdp - SdT + \sum_j \mu_j dn_j$

Progress of reaction $n_j = n_{j0} + v_j \alpha$ or $dn_j = v_j d\alpha$

$$\Rightarrow dG = Vdp - SdT + \sum_j v_j \mu_j d\alpha$$

➤ Equilibrium condition

$$\left(\frac{\partial G}{\partial \alpha} \right)_{T, P} = \sum_j v_j \mu_j = 0$$

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- Perfect gas mixture

$$\mu_j(T, P, \underline{y}) = \mu_{j0}(T, P^*, \underline{y}^*) + RT \ln \frac{Py_j}{P^*}$$

- Non-ideal gas mixture

$$\mu_j(T, P, \underline{y}) = \mu_{j0}(T, P^*, \underline{y}^*) + RT \ln \frac{f_j}{f_j^*}$$

- Solutions

$$\mu_j(T, P, \underline{x}) = \mu_{j0}(T, P^*, \underline{x}^*) + RT \ln \gamma_j x_j$$

$\underline{y}$ = composition ,
 T = temperature,
 P = pressure
 superscript r = reference
 $P^* = 1 \text{ atm}$
 $\underline{y}^*$ = pure j
 f_j = fugacity

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For a reaction

$$V_1 A_1 + V_2 A_2 + V_3 A_3 + V_4 A_4 = 0$$

$$\mu_j = \mu_{j0} + RT \ln(a_j)$$

$$\sum_{j=1}^N V_j \mu_j = \sum_{j=1}^N V_j \mu_{j0} + RT \sum_{j=1}^N V_j \ln(a_j)$$

$$\Delta G = \Delta G^0 + RT \ln \left(\prod_j a_j^{V_j} \right)$$

$$\Delta G = \Delta G^0 + RT \ln K_e$$

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Equilibrium condition

$$V_1 A_1 + V_2 A_2 + V_3 A_3 + V_4 A_4 = 0$$

$$\Delta G = \Delta G^0 + RT \ln \left(\prod_j a_j^{V_j} \right) = 0$$

$$\Delta G = \Delta G^0 + RT \ln K_e = 0$$

$$K_e = \left(\prod_j a_j^{V_j} \right) = \exp \left(\frac{-\Delta G^0}{RT} \right)$$

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Equilibrium constant

$$V_1 A_1 + V_2 A_2 + V_3 A_3 + V_4 A_4 = 0$$

$$K_e = \left(\prod_j a_j^{V_j} \right) = \exp \left(\frac{-\Delta G^0}{RT} \right)$$

$$\text{Pressure} \quad K_e = \left(\prod_j P_j^{V_j} \right) = \left(\prod_j (P_j)^{V_j} \right)$$

$$\text{Fugacity} \quad K_e = \left(\prod_j f_j^{V_j} \right)$$

$$\text{Concentration} \quad K_e = \left(\prod_j c_j^{V_j} \right)$$

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Kinetics and Reactor Design (CHE 7522)

Extent of reaction at equilibrium

$$V_1 A_1 + V_2 A_2 + V_3 A_3 + V_4 A_4 = 0$$

$$K_p = \left(\prod_i P_i^{v_i} \right) \quad P_i = y_i P = \frac{N_i}{N_T} P$$

$$N_j = N_{j0} + v_j \alpha$$

$$K_p = \left(\prod_i \left[P \frac{N_{i0} + v_i \alpha}{N_{T0} + \alpha \sum_j v_j} \right]^{v_i} \right) = F(\alpha)$$

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Kinetics and Reactor Design (CHE 7522)

Operating conditions

$$V_1 A_1 + V_2 A_2 + V_3 A_3 + V_4 A_4 = 0$$

$$K_p(T) = \left(\prod_i P_i \frac{N_{i0} + v_i \alpha}{N_{T0} + \alpha \sum_j v_j} \right) = F(\alpha, P, N_{j0})$$

$$\frac{d}{dT} [\ln K_p(T)] = \frac{d}{dT} [F(\alpha, P, N_{j0})] = \frac{\partial F}{\partial T} + \frac{\partial F}{\partial \alpha} \frac{d\alpha}{dT}$$

$$\frac{d\alpha}{dT} = C \frac{F(\alpha)}{F'(\alpha)}$$

$$\text{Temperature } C = \frac{\Delta H}{RT^2} \quad \Delta H = \text{heat of reaction}$$

$$\text{Pressure } C = -\frac{\sum v_j}{P} \quad \sum v_j = \text{change in no. of moles}$$

Ref: Kenneth Denbigh, The principles of chemical equilibrium

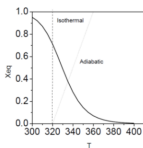
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Kinetics and Reactor Design (CHE 7522)

Reversible-Exothermic reaction



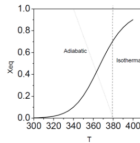
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Kinetics and Reactor Design (CHE 7522)

Reversible-Endothermic reaction



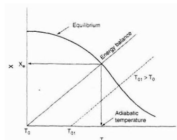
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Kinetics and Reactor Design (CHE F102)

Equilibrium conversion



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Kinetics and Reactor Design (CHE F102)

A = B, first order, Find out the adiabatic temperature and equilibrium temp.

T	K_c	X_c
298	100,000.00	1.00
350	661.60	1.00
400	18.17	0.95
425	4.14	0.90
450	1.11	0.53
475	0.34	0.25
500	0.12	0.11

T (K)	300	400	500	600
X_{eq}	0	0.25	0.50	0.75

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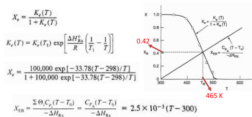
47

Kinetics and Reactor Design (CHE F102)

Adiabatic temp. & Conv.

A = B, feed enters at 300 K (pure A)

Data: $H_{A0} = -40000$ cal/mol, $H_{B0} = -60000$ cal/mol, $C_{pA} = C_{pB} = 50$ cal/mol.K, $K_d(298K) = 10^5$ at 298 K



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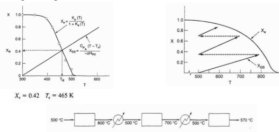
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Kinetics and Reactor Design (CHE F102)

Inter stage Cooling

Higher conversions can be achieved by connecting reactors with inter stage cooling



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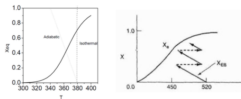
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Kinetics and Reactor Design (CHE 7522)

Endothermic reactions- Inter stage heating

Heat the streams in between



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Kinetics and Reactor Design (CHE 7522)

Heat load & HT area

- What conversion could be achieved in previous problem if two interchange coolers that had the capacity to cool the exit stream to 350 K were available? Also determine the heat duty of each exchanger for a molar feed rate of 40 mol/s. Assume that 95% of equilibrium conversion is achieved in each reactor. The feed temperature to the first reactor is 300 K. (Data: cooling water inlet temperature is 270 K, $c_{pc} = 18 \text{ cal/mol} \cdot \text{K}$, $U = 100 \text{ cal/s} \cdot \text{m}^2 \cdot \text{K}$)

$$X_{E8} = \frac{\sum \Delta C_p (T - T_c)}{-\Delta H_{R2}} = \frac{C_{p2} (T - T_c)}{-\Delta H_{R2}} = 2.5 \times 10^{-3} (T - 300)$$

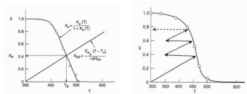
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Kinetics and Reactor Design (CHE 7522)

Solution



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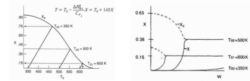
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Kinetics and Reactor Design (CHE 7522)

Optimum feed temperature

- Consider for an adiabatic reactor of fixed size or catalyst weight and investigate What happens if the feed temperature is varied, the FOON is reversible & exothermic
- If Temp is high $\rightarrow$ Rate is high $\rightarrow$ Eq. conv is close to zero $\rightarrow$ little product
- If Temp is low $\rightarrow$ low product (low rate) $\rightarrow$ Eq. conv is high



Along the axial profile: At high temp, the rate is very rapid and equilibrium is achieved very near the reactor entrance (upline). Conversion and temp increase very rapidly over a short distance (small amount of catalyst required)

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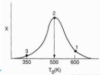
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Kinetics and Reactor Design (CHE 5522)

Optimum feed temperature

- At low and high temp. the conversions are low.
- There must be a optimum temp. that maximizes the conversion
- As the feed T increases from a very low-> rate increases -> the conversion increases with Feed T increases until the Eq. conversion is approached in the reaction.
- Further increase in feed T for exo rxn, decrease the conversion due to the equilibrium conversion.



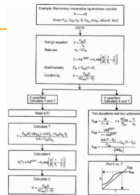
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Kinetics and Reactor Design (CHE 5522)

Algorithm for CSTR Design



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Kinetics and Reactor Design (CHE 5522)

Sizing the non-adiabatic CSTR

Specify X, find V and T

- Calc T $T = T_c + \frac{(-\Delta H_R X)}{C_{p,i}(1 + \epsilon)}$ $\epsilon = \frac{\sum_i \nu_i C_{p,i}}{F_{A0} C_{p,i}}$ and $T_c = \frac{\Delta H_R + T_c}{1 + \epsilon}$
- $K = A \exp(-E/RT)$
- $-r_A = KC_{A0}(1-X)$
- $V = F_{A0} X / -r_A$

Specify T, find X and V

- Calc X $X = \frac{C_{A0}(1 + \epsilon)(T - T_c)}{-\Delta H_R K}$
- $K = A \exp(-E/RT)$
- $-r_A = KC_{A0}(1-X)$
- $V = F_{A0} X / -r_A$

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Kinetics and Reactor Design (CHE 5522)

Problem

- Propylene glycol is produced by the hydrolysis of propylene oxide. Unfortunately, the reactor is beginning to leak, and you must replace it. There is a nice-looking overflow CSTR of 300-gal capacity standing idle: it is glass-lined.

You are feeding 43.04 lb mol/h of propylene oxide (P.O.) to the reactor. The feed stream consists of (1) an equi volumetric mixture of propylene oxide (46.62 ft³/h) and methanol (46.62 ft³/h) and (2) water containing 0.1 wt % H₂SO₄. The volumetric flow rate of water is 233.1 ft³/h, which is 2.5 times the methanol-P.O. flow rate. The corresponding molar feed rates of methanol and water are 71.87 and 802.8 lb mol/h, respectively. The water-propylene oxide-methanol mixture undergoes a slight decrease in volume upon mixing (approximately 3%), but you neglect this decrease in your calculations. The temperature of both feed streams is 58°F prior to mixing, but there is an immediate 17°F temperature rise upon mixing of the two feed streams caused by the heat of mixing. The entering temperature of all feed streams is thus taken to be 75°F.

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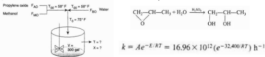
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Kinetics and Reactor Design (CHE 3302)



Furusawa et al state that under conditions similar to those at which you are operating, the reaction is first-order in propylene oxide concentration and apparent zero-order in excess of water with the specific reaction rate



There is an important constraint on your operation. Propylene oxide is a rather low-boiling substance. With the mixture you are using, you feel that you cannot exceed an operating temperature of 125°F , or you will lose too much oxide by vaporization through the vent system.

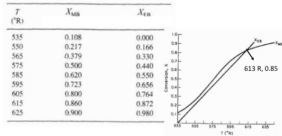
Can you use the idle CSTR as a replacement for the leaking one if it will be operated adiabatically? If so, what will be the conversion of oxide to glycol?

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Kinetics and Reactor Design (CHE 3302)



$125^\circ\text{F} = 585^\circ\text{F}$ We can't use the existing system.

Are there any ways?

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Kinetics and Reactor Design (CHE 3302)



- A cooling coil has been located in equipment storage for use in the hydration of propylene oxide discussed in previous problem. The cooling coil has 40 ft^2 of cooling surface and the cooling water flow rate inside the coil is sufficiently high that a constant coolant temperature of 85°F can be maintained. A typical overall heat-transfer coefficient for such a coil is $100 \text{ Btu/h-ft}^2\text{-}^\circ\text{F}$. Will the reactor satisfy the previous constraint of 125°F maximum temperature if the cooling coil is used?

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Kinetics and Reactor Design (CHE 3302)



PFRR- Heat Removal

- When we checked the vapor pressure at the exit to the adiabatic reactor in the previous problem where the temperature is 360 K , we found the vapor pressure to be about 1.5 MPa for isobutene which is greater than the rupture pressure of the glass vessel being used. Fortunately, there is a bank of ten partially insulated ($U_a = 5000 \text{ kJ/h} \cdot \text{m}^3 \cdot \text{K}$) tubular reactors each 6 m^3 over in the storage shed available for use. We are also going to lower the entering temperature to 305 K . The reactors are cooled by cooling water with flow rate of 500 kg/sec and available at 310 K . The temperature in any of the reactors cannot rise above 325 K . Plot X , X_p , T , and the reaction rate along the length of the reactor. Does the temperature rise above 325 K ?

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Kinetics and Reactor Design (CHE 4022)

Problem

```
function dYfuncecdV = ODEfun(V,Yfuncecd);
Ta = Yfuncecd(1);
T = Yfuncecd(2);
X = Yfuncecd(3);
% Explicit equations
Cao = 9.3;
k = 31.1 * exp(7906 * (T - 360) / (T * 360));
Kc = 3.03 * exp(-830.3 * (T - 333) / (T * 333));
ra = 0 - (k * Cao * (1 - ((1 + 1 / Kc) * X)));
Xe = Kc / (1 + Kc);
CpH = 159;
dH = -6900;
FAo = 14.67;
UA = 5000;
m = 500;
Cpc = 24;
% Differential equations
dTadV = 5000 * (T - Ta) / 14000;
dTdV = (ra * dH - (UA * (T - Ta))) / (CpH * FAo);
dXdV = 0 - (ra / YHoi);
dYfuncecdV = [dTadV; dTdV; dXdV];
```

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Kinetics and Reactor Design (CHE 4022)

Conversion profile

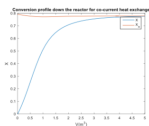
```
clc
Vspan = [0 5]; % Range for the independent variable i.e.volume of the reactor
y0 = [310.; 305.; 0]; % Initial values for the dependent variables Ta, T, and x
[V y]=ode45(@ODEfun,Vspan,y0);
%size(y);
for i=1 : z(1,1)
    %Calculating the values of equilibrium conversion from the following expressions
    Kc(i)=3.03*exp(-830.3*(y(i,2)-333)/(y(i,2)*333));
    Xe(i)=Kc(i)/(1+Kc(i));
end
plot(V,y(:,3),'v','k');
legend('X','K_e');
xlabel('X');
ylabel('V(m^3)');
title('Conversion profile down the reactor for co-current heat exchange');
```

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Kinetics and Reactor Design (CHE 4022)



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Kinetics and Reactor Design (CHE 4022)

Rate profile

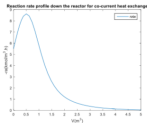
```
clc
Vspan = [0 5]; % Range for the independent variable i.e.volume of reactor
y0 = [310.; 305.; 0]; % Initial values for the dependent variables and x
[V y]=ode45(@ODEfun,Vspan,y0);
%size(y);
for i=1 : z(1,1)
    %Calculating the values of rate from the following expressions
    Kc(i)=3.03*exp(-830.3*(y(i,2)-333)/(y(i,2)*333));
    K(i)=31.1*exp(7906*(y(i,2)-360)/(y(i,2)*360));
    ra(i)=-K(i)*9.3*(1-(1+1/Kc(i))*y(i,3));
end
plot(V,-ra);
xlabel('-ra(kmol/m^3.h)');
ylabel('V(m^3)');
axis([0 5 0 9]);
legend('rate');
title('Reaction rate profile down the reactor for co-current heat exchange');
```

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Kinetics and Reactor Design (ENH F532)



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Kinetics and Reactor Design (ENH F532)

Temperature profile

```

clc
Vspan = [0 5]; % Range for the independent variable
i.e., volume of the reactor
y0 = [310.; 305.; 0]; % Initial values for the
dependent variables Ta, T, and x

[v y]=ode45(@ODEfun,Vspan,y0);

plot(v,y(:,1),v,y(:,2));
legend('T', 'a', 'T');
ylabel('T(K)');
xlabel('V(m^3)');
axis([0 5 300 320]);
title('Temp profile down the reactor for co-current
heat exchange')

```

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Kinetics and Reactor Design (ENH F532)

Problem

- Propylene glycol is produced by the hydrolysis of propylene oxide. Unfortunately, the reactor is beginning to leak, and you must replace it. There is a nice-looking overflow CSTR of 300-gal capacity standing idle: it is glass-lined. You are feeding 43.04 lb mol/h of propylene oxide (P.O.) to the reactor. The feed stream consists of (1) an equi volumetric mixture of propylene oxide (46.62 ft³/h) and methanol (46.62 ft³/h) and (2) water containing 0.1 wt % H₂SO₄. The volumetric flow rate of water is 233.1 ft³/h, which is 2.5 times the methanol-PO. flow rate. The corresponding molar feed rates of methanol and water are 71.87 and 802.8 lb mol/h, respectively. The water-propylene oxide-methanol mixture undergoes a slight decrease in volume upon mixing (approximately 3%), but you neglect this decrease in your calculations. The temperature of both feed streams is 58°F prior to mixing, but there is an immediate 17°F temperature rise upon mixing of the two feed streams caused by the heat of mixing. The entering temperature of all feed streams is thus taken to be 75°F.

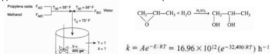
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Kinetics and Reactor Design (ENH F532)

Furusawa et al state that under conditions similar to those at which you are operating, the reaction is first-order in propylene oxide concentration and apparent zero-order in excess of water with the specific reaction rate



There is an important constraint on your operation. Propylene oxide is a rather low-boiling substance. With the mixture you are using, you feel that you cannot exceed an operating temperature of 125°F, or you will lose too much oxide by vaporization through the vent system. Can you use the idle CSTR as a replacement for the leaking one if it will be operated adiabatically? If so, what will be the conversion of oxide to glycol?

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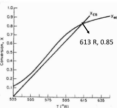
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Kinetics and Reactor Design (CHE 5522)



T (°R)	X_{sub}	X_{eq}
535	0.108	0.000
550	0.217	0.166
565	0.379	0.330
575	0.500	0.440
585	0.620	0.550
595	0.723	0.656
605	0.800	0.764
615	0.860	0.872
625	0.900	0.980



125°F = 585°F We can't use the existing system.

Are there any ways?

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Kinetics and Reactor Design (CHE 5522)



- A cooling coil has been located in equipment storage for use in the hydration of propylene oxide discussed in previous problem. The cooling coil has 40 ft² of cooling surface and the cooling water flow rate inside the coil is sufficiently high that a constant coolant temperature of 85°F can be maintained. A typical overall heat-transfer coefficient for such a coil is 100 Btu/h-ft² °F. Will the reactor satisfy the previous constraint of 125°F maximum temperature if the cooling coil is used?

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Kinetics and Reactor Design (CHE 5522)

CSTR with cooling jacket : Eg.8.9



```
function fx = NONLEfun(x);
X = x(1);
T = x(2);
% Explicit equations
A = 16.96 * 10 ^ 12;
R = 1.987;
E = 32400;
k = A * exp(0 - (E / (R * T)));
tau = 0.1229;
% Nonlinear equations
fx(1,1) = X - ((403.3 * (T - 535) + 92.9 * (T - 545)) /
(36400 + 7 * (T - 528)));
fx(2,1) = X - (tau * k / (1 + tau * k));
```

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Kinetics and Reactor Design (CHE 5522)



```
clc
xguess = [0.367 564.]; % initial guess vector
for X and T
y = fsolve(NONLEfun,xguess);
X=y(1);
T=y(2);
disp('The exiting temperature(degree R) is')
T
disp('The exiting conversion is')
X
```

T = 563.7289

The exiting conversion is X = 0.3636

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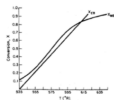
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Kinetics and Reactor Design (CHE 7522)

Specify V , find X and T Use * X eqn. generate T vs X

$$X = \frac{C_{p0}(1 + \kappa)(T - T_c)}{-\Delta H^0_{Rx}}$$

 $\Rightarrow V = F_{A0} X / -r_A$ $\Rightarrow X_{MB} = \tau k / (1 + \tau k)$ $\Rightarrow$ Generate T vs X_{MB} $\Rightarrow$ Plot T vs X_{MB} and X_{MB} 

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Kinetics and Reactor Design (CHE 7522)



Multiple steady states- CSTR

$$\frac{G(T)}{-X \Delta H^0_{Rx}} = \frac{R(T)}{C_{p0}(1 + \kappa)(T - T_c)}$$

$$X = \frac{C_{p0}(1 + \kappa)(T - T_c)}{-\Delta H^0_{Rx}}$$

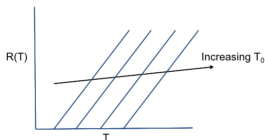
$$T = T_c + \frac{(-\Delta H^0_{Rx})(X)}{C_{p0}(1 + \kappa)}$$

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Kinetics and Reactor Design (CHE 7522)

Effect of T 

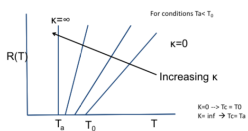
Variation of heat removal line with inlet temperature.

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Kinetics and Reactor Design (CHE 7522)

Effect of κ Variation of heat removal line with κ ($\kappa = UA/C_{p0}F_{A0}$)

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Kinetics and Reactor Design (CHE F102)



$$V = \frac{F_{A0}X}{-r_A(X, T)}$$



1) Mole Balances: $V = \frac{F_{A0}X}{-r_A}$

2) Rate Laws: $-r_A = kC_A$

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Kinetics and Reactor Design (CHE F102)



3) Stoichiometry: $C_A = C_{A0}(1 - X)$

4) Combine: $V = \frac{F_{A0}X}{kC_{A0}(1 - X)} = \frac{C_{A0}X}{kC_{A0}(1 - X)}$

$$\tau k = \frac{X}{1 - X}$$

$$X = \frac{\tau k}{1 + \tau k} = \frac{\tau A e^{-E/RT}}{1 + \tau A e^{-E/RT}}$$

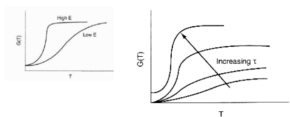
$$G(T) = X(-\Delta H_{Rx}) = \frac{\tau A e^{-E/RT}}{1 + \tau A e^{-E/RT}} (-\Delta H_{Rx})$$

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Kinetics and Reactor Design (CHE F102)



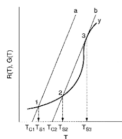
Variation of heat generation curve with space-time.

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Kinetics and Reactor Design (CHE F102)



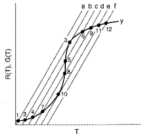
Finding Multiple Steady States with T_0 varied

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Kinetics and Reactor Design (CHE 5302)



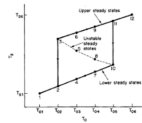
Finding Multiple Steady States with T_0 varied

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Kinetics and Reactor Design (CHE 5302)



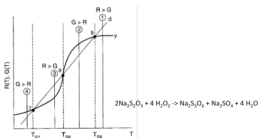
Temperature ignition-extinction curve

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Kinetics and Reactor Design (CHE 5302)



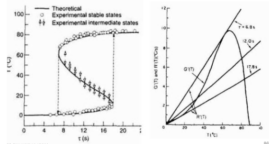
Stability of multiple state temperatures

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Kinetics and Reactor Design (CHE 5302)



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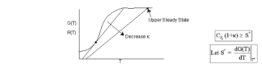
Kinetics and Reactor Design (CHE 3322)

Runaway Reactions in CSTRs

- Runaway reaction - when it goes to the upper steady state which is at an unacceptably high temperature, such as the propylene glycol
- when the point of tangency between R(T) and Q(T) disappears and only the upper steady state exists-> Runaway occurs
- If heat exchange term UA is decreased. As a result k will decrease and we will no longer have a point of tangency and as a result the CSTR will **runaway** to the upper steady state.



Kinetics and Reactor Design (CHE 3322)



As a result, this decrease in κ will cause $\left. \frac{dT}{dt} \right|_{T_c} < 0$ and we have a runaway.

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Kinetics and Reactor Design (CHE 3322)

Stability

$$\Delta T_{ex} = T^* - T_c = \frac{RT_c^2}{E}$$

If this difference between the reactor temp. and T_c , ΔT_{ex} , is exceeded, transition to upper steady state will occur.

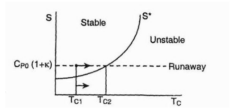
$$T^* = \frac{E}{2R} \left[1 + \sqrt{1 - 4 \frac{F R}{E}} \right] \quad T_c = \frac{u T^* + T_c}{1 + \kappa}$$

For many industrial reactions, E/RT is typically between 16 to 24, the rxn temp. may be around 300 to 500 K, then, ΔT_{ex} = 15 to 30 °C

Vary T_c , then calculate T^* , κ^* , κA^* , $Q(T^*)$ and finally $S^* = (E/RT_c^2) Q(T^*)$. Any deviation to the right or below the intersection of $C_{p0}(1+\kappa)$ and S^* will result in runaway.



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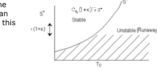
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Kinetics and Reactor Design (CHE 5522)



for example, $C_{A1}(1+\kappa) = 30$ then to the left of T_{C1} , and the reactor is stable, so we can increase either T_0 or T_c in this region up to T_{C1}



if we decrease to $C_{A1}(1+\kappa) = 10$, then we see we have a narrower operating range over which T_0 and T_a may be increased.



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Kinetics and Reactor Design (CHE 5522)

CSTR Runaway For Zero and First Order Reactions



- The isomerization reaction is carried out in a CSTR with a heat exchanger. Construct a stability diagram.

The ratio of inerts to A fed is 2 to 1.

$$\begin{aligned}\tau &= 0.1 \text{ min} \\ A_0 &= 1.5 \times 10^7 \text{ mol/dm}^3 \cdot \text{min}^{-1} \\ A_1 &= 1.5 \times 10^7 \text{ min}^{-1} \\ E &= 10,000 \text{ cal/mol} \\ R &= 1.987 \text{ cal/mol/K} \\ U_A &= 600 \text{ cal/min/dm}^2 \cdot \text{K} \\ C_{P_A} &= 10 \text{ cal/mol/K}\end{aligned}$$

$$\begin{aligned}C_{P_1} &= 20 \text{ cal/mol/K} \\ T_c &= \frac{T_0 + \kappa T_A}{1 + \kappa} = T_0 \\ T_0 &= T_A \\ C_{P_2} &= \sum C_{P_i} = 10 + 2(10) = 50 \text{ cal/mol/K} \\ C_{A0} &= 2.0 \text{ mol/dm}^3 \\ \Delta H_{R2} &= -10,000 \text{ cal/mol}\end{aligned}$$

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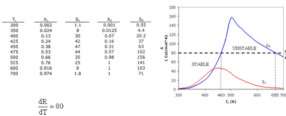
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Runaway- Zero and First order



$$\frac{dT_c}{dT_c} = 0$$

The first order reaction is globally stable for all T_c . However, we see the zero order reaction will run away when the entering temperature T_c is increased above 460K. We also note the runaway reaction becomes stable again as T_c is further increased to values above 685K

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Kinetics and Reactor Design (CHE 5522)



Multiple Reactions – PFR/ PBR

PFR/PBR:

$$\frac{dT}{dV} = \frac{Ua(T_s - T) + (-r_A)(-\Delta H_{R1}(T))}{\sum_{j=1}^N F_j C_{Pj}}$$

$$\frac{dT}{dV} = \frac{Ua(T_s - T) + \sum_{j=1}^N (-r_j)(\Delta H_{Rj})}{\sum_{j=1}^N F_j C_{Pj}}$$

These equations are coupled with the mole balances and rate law equations.

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Parallel Reactions in a PFR with Heat Effects



Pure A is fed at a rate of 100 mol/s, a temperature of 150 °C, and a concentration of 0.1 mol/dm³. Determine the temperature and flow rate profiles down the reactor.

$$\Delta H_{R1,A} = -20,000 \text{ J/mol of A reacted in reaction 1)}$$

$$\Delta H_{R2,A} = -60,000 \text{ J/mol of A reacted in reaction 2)}$$

$C_{p,A} = 90 \text{ J/mol} \cdot ^\circ\text{C}$	$k_{1,150} = 10 \exp\left[\frac{E_1}{R} \left(\frac{1}{300} - \frac{1}{T}\right)\right] \text{ s}^{-1}$
$C_{p,R} = 90 \text{ J/mol} \cdot ^\circ\text{C}$	$E_1/R = 4000 \text{ K}$
$C_{p,C} = 180 \text{ J/mol} \cdot ^\circ\text{C}$	$k_{2,150} = 0.09 \exp\left[\frac{E_2}{R} \left(\frac{1}{300} - \frac{1}{T}\right)\right] \frac{\text{dm}^3}{\text{mol} \cdot \text{s}}$
$Ua = 4000 \text{ J/m}^2 \cdot \text{s} \cdot ^\circ\text{C}$	$E_2/R = 9000 \text{ K}$
$T_a = 100^\circ\text{C}$	

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